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Bent Connector, Intermediate Connector, and Connector Assembly

Abstract

A bent connector has an outer case including a case body, an inner insertion groove recessed in a first direction from a first side of the case body toward an inside of the case body, and a terminal insertion groove recessed in a second direction from a second side of the case body toward the inside of the case body. The bent connector has an inner housing inserted through the inner insertion groove, a wire assembly having at least a portion inserted into an inside of the inner housing, a terminal bus bar inserted through the terminal insertion groove and electrically connected to the wire assembly, and a fastening bolt physically and electrically fastening the terminal bus bar and the wire assembly to each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of the filing date under 35 U.S.C. § 119(a)-(d) of Korean Patent Application No. 10-2024-0031510, filed on Mar. 5, 2024, and Korean Patent Application No. 10-2025-0016521, filed on Feb. 10, 2025.

FIELD OF THE INVENTION

[0002] One or more embodiments relate to a bent connector, an intermediate connector, and a connector assembly.

BACKGROUND OF THE INVENTION

[0003] A connector is a type of electrical component that may connect or disconnect electrical connections. The connector is used in various electronic and mechanical devices, such as automobiles or home appliances, to electrically and/or mechanically connect a plurality of electronic components.

[0004] Technologies are required to prevent damage to connection terminals by assisting in the arrangement of the connection terminals of the connector. In addition, technologies are required to manufacture the connector in a compact size.

[0005] The above description is information the inventor(s) acquired during the course of conceiving the present disclosure, or already possessed at the time, and is not necessarily art publicly known before the present application was filed.

SUMMARY OF THE INVENTION

[0006] A bent connector has an outer case including a case body, an inner insertion groove recessed in a first direction from a first side of the case body toward an inside of the case body, and a terminal insertion groove recessed in a second direction from a second side of the case body toward the inside of the case body. The bent connector has an inner housing formed of an insulating material and inserted through the inner insertion groove from the first side of the case body, a wire assembly having at least a portion inserted into an inside of the inner housing in a direction in which the inner housing is inserted into the inner insertion groove, a terminal bus bar inserted through the terminal insertion groove from the second side of the case body and electrically connected to the wire assembly, and a fastening bolt inserted from the second side of the case body and mechanically and electrically fastening the terminal bus bar and the wire assembly to each other.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The invention will now be described by way of example with reference to the accompanying figures, of which:

[0008] FIG. 1 is a perspective view illustrating a connector assembly installed in an electronic device, according to an embodiment;

[0009] FIG. 2 is a perspective view illustrating a bent connector and an intermediate connector that are separated from the electronic device, according to an embodiment;

[0010] FIG. 3 is an exploded cross-sectional view of the connector assembly according to an embodiment;

[0011] FIG. 4 is a cross-sectional view illustrating a portion of the bent connector exploded, according to an embodiment;
[0012] FIG. 5 is a perspective view illustrating a coupling relationship between a terminal housing and a terminal bus bar, according to an embodiment;
[0013] FIG. 6 is a cross-sectional view and a partially enlarged view of FIG. 5;
[0014] FIG. 7 is a perspective view of the intermediate connector according to an embodiment;
[0015] FIG. 8 is a cross-sectional view of the intermediate connector according to an embodiment;
and
[0016] FIG. 9 is a perspective view illustrating a portion of a bent connector according to an embodiment.

DETAILED DESCRIPTION

[0017] Hereinafter, embodiments are described in detail with reference to the accompanying drawings. However, various alterations and modifications may be made to the embodiments and thus, the scope of the disclosure is not limited or restricted to the embodiments. The embodiments should be understood to include all changes, equivalents, and replacements within the idea and the technical scope of the disclosure.

[0018] The terminology used herein is for the purpose of describing particular embodiments only and is not to be limiting of the embodiments. The singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises/comprising” and/or “includes/including” when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components and/or groups thereof.

[0019] As used herein, “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B or C”, “at least one of A, B and C”, and “at least one of A, B, or C,” each of which may include any one of the items listed together in the corresponding one of the phrases, or all possible combinations thereof.

[0020] Unless otherwise defined, all terms including technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the embodiments belong. It will be further understood that terms, such as those defined in commonly-used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0021] When describing the embodiments with reference to the accompanying drawings, like reference numerals refer to like constituent elements and a repeated description related thereto will be omitted. In the description of embodiments, detailed description of well-known related structures or functions will be omitted when it is deemed that such description will cause ambiguous interpretation of the present disclosure.

[0022] Also, in the description of the components, terms such as first, second, A, B, (a), (b) or the like may be used herein when describing components of the present disclosure. These terms are used only for the purpose of discriminating one component from another component, and the nature, the sequences, or the orders of the components are not limited by the terms. When one component is described as being “connected”, “coupled”, or “attached” to another component, it should be understood that one component may be connected or attached directly to another component, and an intervening component may also be “connected”, “coupled”, or “attached” to the components.

[0023] Referring to FIGS. 1 to 4, a connector assembly 1 according to an embodiment may be connected to an electronic device 2 (e.g., a power supply, a battery module, and/or an inverter) provided inside a vehicle, for example, to transmit power from the electronic device 2 to the outside or from the outside to the electronic device 2. Here, the electronic device 2 may include a

device case **21** and a device bus bar **23** electrically connected to an internal component of the device case **21**. For example, the device bus bar **23** may be accommodated inside an intermediate groove **22** formed to be recessed from an outer surface of the device case **21**, but embodiments are not limited thereto.

[0024] According to an embodiment, the connector assembly **1** may be designed such that a height protruding from an outer surface of the electronic device **2** is low, thereby reducing overall volume. In this structure, the space that the electronic device **2** may occupy in the vehicle may be increased. For example, a width direction (e.g., a $\pm$ -x-axis direction) of the connector assembly **1** may be parallel to an outer surface (e.g., an upper surface (a surface perpendicular to a +z-axis) and/or a front surface (a surface perpendicular to a +y-axis) with respect to FIG. 1) of the electronic device **2** with which the connector assembly **1** is in contact. For example, the connector assembly **1** may include a bent connector **11** and an intermediate connector **12**.

[0025] According to an embodiment, the bent connector **11** may be installed on the outer surface of the electronic device **2**. For example, the bent connector **11** may receive power from a conductive member (e.g., the device bus bar **23**) formed in a vertical direction (e.g., a +z-axis direction) from the inside of the electronic device **2** and may bend transmission direction of the power to transmit the power in a direction parallel to the outer surface (e.g., the upper surface with respect to FIG. 1) of the electronic device **2**. For example, the bent connector **11** may include an outer case **111**, an inner housing **112**, a fastening nut **113**, a wire assembly **114**, a terminal support housing **115**, a terminal bus bar **116**, and a fastening bolt **117**.

[0026] According to an embodiment, as shown in FIG. 4, the outer case **111** may include a case body **1111**, an inner insertion groove **1112** that is formed to be recessed in a direction (e.g., a +y-axis direction) from one side (e.g., an end side in a -y-axis direction) of the case body **1111** toward the inside of the case body **1111**, a terminal insertion groove **1113** that is formed to be recessed in a direction (e.g., the +z-axis direction) from the other side (e.g., an end side in a -z-axis direction) of the case body **1111** toward the inside of the case body **1111** in a direction (e.g., a $\pm$ -z-axis direction) intersecting with a direction (e.g., the +y-axis direction) in which the inner insertion groove **1112** is recessed, and an outer-side inner locking structure **1114** in which at least a portion of the inner housing **112** may be caught.

[0027] According to an embodiment, the case body **1111** may be formed of a conductive material. With this configuration, a ground line may be formed to transmit current leaking from the wire assembly **114** to the outside through a grounding ferrule **1145** of the wire assembly **114** and the case body **1111**. Therefore, it may be possible to reduce issues of causing operational errors in electronic components, such as leakage current acting as noise on electronic components inside the vehicle.

[0028] According to an embodiment, the inner insertion groove **1112** and the terminal insertion groove **1113** may communicate with each other. In a portion where the inner insertion groove **1112** and the terminal insertion groove **1113** communicate with each other, the wire assembly **114** and the terminal bus bar **116** may be connected to each other.

[0029] According to an embodiment, the outer-side inner locking structure **1114** may be formed to be recessed on an inner wall of the case body **1111**. In this structure, a portion (e.g., an inner-side outer locking structure **1122**) of the inner housing **112** is caught by the outer-side inner locking structure **1114**, as shown in FIG. 4, inhibiting the inner housing **112** from being separated from the outer case **111**. In another example, the inner-side outer locking structure **1122** may be caught in a corner area that is in contact with the inner insertion groove **1112**, in the terminal insertion groove **1113**. In this case, it may be understood that the above-described corner area is the outer-side inner locking structure **1114**. In this structure, since there is no need to form a separate groove on the inner wall of the case body **1111**, manufacturing cost of the case body **1111** may be reduced.

[0030] According to an embodiment, the inner housing **112** may be inserted through the inner insertion groove **1112** from one side (e.g., the end side in the -y-axis direction) of the case body

1111. For example, the inner housing **112** may include an inner body **1121**, the inner-side outer locking structure **1122**, a wire insertion groove **1123**, a ferrule insertion groove **1124**, a plate insertion groove **1125**, and a nut groove **1126**.

[0031] According to an embodiment, the inner body **1121** may be formed of an insulating material (e.g., resin) to reduce power from being directly transmitted from a conductive wire **1141** or a conductive plate **1146** of the wire assembly **114** to the outer case **111**. For example, at least a portion of the inner body **1121** may be positioned at a portion where the inner insertion groove **1112** and the terminal insertion groove **1113** meet each other.

[0032] According to an embodiment, the inner-side outer locking structure **1122** may have a shape that protrudes outwardly (e.g., in the +z-axis direction) from the inner housing **112** as the inner-side outer locking structure **1122** extends in a direction (e.g., the -y-axis direction) where the inner housing **112** is separated from the inner insertion groove **1112**. In this structure, when the inner housing **112** is inserted into the inner insertion groove **1112** by a predetermined length or more, the inner-side outer locking structure **1122** is caught on an inner wall (e.g., the inner-side outer locking structure **1122**) of the outer case **111**, thereby inhibiting the inner housing **112** from being separated from the inner insertion groove **1112**.

[0033] According to an embodiment, the wire insertion groove **1123** may be formed to be recessed in a direction (e.g., the +y-axis direction) in which the inner insertion groove **1112** is recessed from one side (right side of FIG. 4) of the inner body **1121**. For example, the wire assembly **114** may be inserted into the wire insertion groove **1123**. For example, between the inner insertion groove **1112** and the inner wall of the case body **1111**, there may be a structure in which the inner insertion groove **1112** and the inner wall of the case body **1111** are surrounded by the inner body **1121**, thereby reducing direct contact between the conductive wire **1141** or the conductive plate **1146** of the wire assembly **114** and the inner wall of the case body **1111**.

[0034] According to an embodiment, the ferrule insertion groove **1124** may be formed to be recessed stepwise from an outer surface of the inner body **1121**. In this structure, at least a portion of the grounding ferrule **1145** of the wire assembly **114** may be inserted into a space between the inner wall of the case body **1111** and the ferrule insertion groove **1124**. For example, an inner surface of the grounding ferrule **1145** may be supported by an outer surface of the ferrule insertion groove **1124**, thereby increasing a contact force between the grounding ferrule **1145** and the inner wall of the case body **1111**.

[0035] According to an embodiment, the plate insertion groove **1125** may be formed to be further recessed in a direction in which the inner insertion groove **1112** is recessed from the wire insertion groove **1123**. For example, the conductive plate **1146** of the wire assembly **114** may be inserted into the plate insertion groove **1125**. For example, at least a portion of an inner wall of the plate insertion groove **1125** may support one surface of the conductive plate **1146**, thereby reducing an issue of the conductive plate **1146** being pushed out during a bolting process using the fastening bolt **117** and allowing the conductive plate **1146** and the terminal bus bar **116** to be stably bolted together.

[0036] According to an embodiment, the nut groove **1126** may be formed to be recessed in a direction (e.g., the +z-axis direction) in which the fastening bolt **117** is inserted from the plate insertion groove **1125** into the case body **1111**. For example, the fastening nut **113** and the nut groove **1126** may have polygonal shapes (e.g., square shapes) that are complementary to each other. According to such a shape, an issue of the fastening nut **113** idling with respect to the nut groove **1126** during the bolting process may be reduced, and thus, a bolting operation may be stably performed. However, it should be noted that the shape of the fastening nut **113** is not limited thereto.

[0037] According to an embodiment, as shown in FIGS. 3 and 4, the fastening nut **113** may be arranged on the opposite side of the terminal bus bar **116** with respect to the conductive plate **1146** of the wire assembly **114** and may be fastened to the fastening bolt **117**. In this structure, the

conductive plate **1146** and the terminal bus bar **116** may be arranged to be in close contact with each other by bolting the fastening nut **113** and the fastening bolt **117** to each other. For example, the fastening nut **113** may be formed of a material (e.g., metal) that is stronger than the inner housing **112**. In this structure, compared to the case where the fastening bolt **117** is directly fastened to the inner housing **112** without the fastening nut **113**, fastening rigidity may be improved and an issue of damage to a screw groove may be reduced, thereby helping to improve a lifespan of an entire product.

[0038] Furthermore, the fastening nut **113** according to an embodiment may be fixed to the conductive plate **1146**, as shown in FIG. 3. For example, the fastening nut **113** may be fixed to a hole formed in the conductive plate **1146** in a press fit and/or welding method, but embodiments are not limited thereto.

[0039] According to an embodiment, the wire assembly **114** may be inserted from one side (e.g., an end side in the $-y$ -axis direction) of the outer case **111**. For example, the wire assembly **114** may be inserted along a direction (e.g., the $+y$ -axis direction) in which the inner housing **112** is inserted into the inner insertion groove **1112**. For example, at least a portion of the wire assembly **114** may be inserted into the inside of the inner housing **112**. For example, the wire assembly **114** may include the conductive wire **1141** formed of a conductive material and inserted along the wire insertion groove **1123**, an inner insulation layer **1142** wrapping the conductive wire **1141**, an outer insulation layer **1144** wrapping the inner insulation layer **1142**, a shield layer **1143** at least partly positioned between the inner insulation layer **1142** and the outer insulation layer **1144**, the grounding ferrule **1145**, the conductive plate **1146**, a wire seal **1147** for reducing foreign substances from entering the inner insertion groove **1112** from the outside, and a wire-side cover **1148** for covering the inner insertion groove **1112**.

[0040] According to an embodiment, the shield layer **1143** is formed of a conductive material and may transmit current leaking from the conductive wire **1141** to the outside (e.g., the case body **1111**) of the wire assembly **114**. For example, the shield layer **1143** may have a mesh shape but is not limited thereto. For example, at least a portion of the shield layer **1143** may be positioned between the inner insulation layer **1142** and the outer insulation layer **1144**. For example, as shown in FIG. 3, the remaining portion of the shield layer **1143** may be in surface contact with the grounding ferrule **1145**. In this structure, the shield layer **1143** may transmit current leaking from the conductive wire **1141** to the grounding ferrule **1145**.

[0041] According to an embodiment, one side of the grounding ferrule **1145** may be in contact with the shield layer **1143**, and the other side of the grounding ferrule **1145** may be in contact with the case body **1111**. For example, the grounding ferrule **1145** may include a ferrule body **1145a** and a contact portion **1145b** that is formed to protrude from the ferrule body **1145a** toward the inner wall of the case body **1111** and is elastically deformable, as shown in FIG. 4.

[0042] According to an embodiment, the conductive plate **1146** may be connected to an end of the conductive wire **1141** and may be inserted along the plate insertion groove **1125**. For example, a width direction (e.g., the $\pm x$ -axis direction) of the conductive plate **1146** may be parallel to a width direction (e.g., the $\pm x$ -axis direction) of the outer case **111**. For example, a normal direction (e.g., the $\pm z$ -axis direction) to a surface of the conductive plate **1146** may be parallel to a direction (e.g., the $+z$ -axis direction) in which the terminal insertion groove **1113** is formed to be recessed. In this structure, in the case where a surface shape of the conductive plate **1146** is arranged in a direction (e.g., the $\pm z$ -axis direction) perpendicular to the outer surface (e.g., an upper surface (the surface perpendicular to the $+z$ -axis) with respect to FIG. 3) of the electronic device **2**, the height of an entire shape of the conductive plate **1146** and the bent connector **11** including the conductive plate **1146** protruding from the outer surface (e.g., the upper surface with respect to FIG. 3) of the electronic device **2** may be reduced. For example, the width direction (e.g., the $\pm x$ -axis direction) of the conductive plate **1146** and a width direction (e.g., the $\pm x$ -axis direction) of the terminal bus bar **116** may each be parallel to a width direction (e.g., the $\pm x$ -axis

direction) of the outer case **111**.

[0043] According to an embodiment, the terminal support housing **115** may be inserted through the terminal insertion groove **1113** from the other side (e.g., a lower side of FIG. 3) of the case body **1111** and formed of an insulating material (e.g., resin). However, it should be noted that, unlike the drawing, the terminal support housing **115** may be omitted. An example of the terminal support housing **115** is described below.

[0044] According to an embodiment, the terminal bus bar **116** may be electrically connected to the wire assembly **114**. For example, the terminal bus bar **116** may be inserted from the other side (e.g., the end side in the $-z$ -axis direction) of the outer case **111**. For example, the terminal bus bar **116** may be inserted through the terminal insertion groove **1113** from the other side of the case body **1111** and electrically connected to the wire assembly **114**. For example, a width direction (e.g., the $+/-x$ -axis direction) of the terminal bus bar **116** may be parallel to the outer surface (e.g., the upper surface (the surface perpendicular to the $+z$ -axis) and/or the front surface (the surface perpendicular to the $+y$ -axis) with respect to FIG. 1) of the electronic device **2**. In this structure, compared to a case where a surface shape of the terminal bus bar **116** is arranged in a direction (e.g., a $+/-y$ -axis direction) perpendicular to the outer surface (e.g., a left surface (a surface perpendicular to the $+y$ -axis) with respect to FIG. 3) of the electronic device **2**, the height of an entire shape of the terminal bus bar **116** and the bent connector **11** including the terminal bus bar **116** protruding from the outer surface (e.g., the left surface with respect to FIG. 3) of the electronic device **2** may be reduced. For example, the width direction (e.g., the $+/-x$ -axis direction) of the terminal bus bar **116** may be parallel to the width direction (e.g., the $+/-x$ -axis direction) of the outer case **111**. For example, an end of the terminal bus bar **116** may have a bent shape with respect to a longitudinal direction (e.g., the $+/-y$ -axis direction) of the wire assembly **114**. For example, one end of the terminal bus bar **116** may be connected to the conductive plate **1146**, and the other end of the terminal bus bar **116** may be extended in a bent shape in a direction (e.g., the $+/-z$ -axis direction) perpendicular to the longitudinal direction (e.g., the $+/-y$ -axis direction) of the wire assembly **114**. An example of the terminal bus bar **116** is described below.

[0045] According to an embodiment, the fastening bolt **117** may be inserted from the other side (e.g., a lower side of FIG. 3) of the outer case **111** (e.g., the case body **1111**) to mechanically and electrically fasten the terminal bus bar **116** and the wire assembly **114** (e.g., the conductive plate **1146**) to each other. For example, the fastening bolt **117** may be fastened to the fastening nut **113** and/or the inner body **1121** by penetrating the terminal bus bar **116** and the conductive plate **1146**.

[0046] According to an embodiment, the intermediate connector **12** may mechanically and electrically connect the electronic device **2** (e.g., the device bus bar **23**) and the bent connector **11** (e.g., the terminal bus bar **116**) to each other. For example, the intermediate connector **12** may include a bidirectional terminal **121**, a device-side intermediate housing **122**, and a connector-side intermediate housing **123**.

[0047] According to an embodiment, as shown in FIG. 3, each end of the bidirectional terminal **121** may be connected to a pair of bus bars (e.g., the device bus bar **23** and the terminal bus bar **116**). In this configuration, in a process of installing the bent connector **11** in the electronic device **2**, separate wiring work may be omitted or reduced, and thus, the time and effort required for an assembly process may be reduced, thereby improving work convenience of a worker. For example, the bidirectional terminal **121** may be formed in a thin plate shape, and a plurality of the bidirectional terminals **121** may be arranged along the width direction (e.g., the $+/-x$ -axis direction) of the terminal bus bar **116**. In this configuration, the number of the bidirectional terminals **121** may be adjusted according to the width of the terminal bus bar **116** or the size of power transmitted through the terminal bus bar **116**. Examples of configurations of other intermediate connectors **12** are described below.

[0048] Referring to FIGS. 4, 5, and 6, the terminal bus bar **116** according to an embodiment may include a vertical portion **1161** and a horizontal portion **1162**.

[0049] According to an embodiment, the vertical portion **1161** may extend parallel to a longitudinal direction (e.g., the $\pm$ -z-axis direction) of the terminal insertion groove **1113**. For example, the vertical portion **1161** may have a shape parallel to the outer surface (e.g., the left surface with respect to FIG. 3) of the electronic device **2**. For example, a normal direction (e.g., the $\pm$ -y-axis direction) to a surface of the vertical portion **1161** may be parallel to a direction (e.g., the $\pm$ -y-axis direction) in which the inner insertion groove **1112** is formed to be recessed. According to an embodiment, the vertical portion **1161** may include at least one fixing protrusion **1161a** that is formed to protrude in a direction perpendicular to a longitudinal direction (e.g., the $\pm$ -z-axis direction) of the vertical portion **1161**. The fixing protrusion **1161a** according to an embodiment may be interference-fitted into an inner wall of a bus bar insertion hole **1152** of the terminal support housing **115**. In this structure, a fastening force between the terminal support housing **115** and the terminal bus bar **116** may be improved, and thus, an issue of the terminal bus bar **116** being separated from the terminal support housing **115** or being inappropriately arranged may be reduced during the assembly process.

[0050] According to an embodiment, the horizontal portion **1162** may have a bent shape from the vertical portion **1161**, may extend parallel to a longitudinal direction (e.g., the $\pm$ -y-axis direction) of the inner insertion groove **1112**, and may be in surface contact with the conductive plate **1146**. For example, the horizontal portion **1162** may have a shape that is parallel to the outer surface (e.g., the upper surface with respect to FIG. 3) of the electronic device **2**.

[0051] According to an embodiment, the terminal support housing **115** may include a support housing body **1151** having a shape corresponding to the terminal insertion groove **1113**, the bus bar insertion hole **1152** into which the vertical portion **1161** is inserted, a bus bar mounting groove **1153** into which the horizontal portion **1162** is mounted, a bolting hole **1154** communicating with the bus bar mounting groove **1153** and through which the fastening bolt **117** may pass, and a terminal-side outer locking structure **1155** on which at least a portion of the inner housing **112** may be caught.

[0052] According to an embodiment, the bus bar insertion hole **1152** may be formed to penetrate the support housing body **1151**. For example, the bus bar insertion hole **1152** may be formed in a shape corresponding to the vertical portion **1161** so that the vertical portion **1161** may be stably accommodated inside the bus bar insertion hole **1152**. For example, the bus bar insertion hole **1152** may be formed of a material having lower rigidity than the fixing protrusion **1161a**, and thus, the fixing protrusion **1161a** may be interference-fitted into an inner wall of the bus bar insertion hole **1152**.

[0053] According to an embodiment, the bus bar mounting groove **1153** may have a stepwise shape from one surface of the support housing body **1151**. For example, the bus bar mounting groove **1153** may have a shape communicating with the bus bar insertion hole **1152**. For example, a partition wall may be provided on at least a portion or all of a circumference of the bus bar mounting groove **1153** so that the horizontal portion **1162** may be stably accommodated in the bus bar mounting groove **1153**.

[0054] According to an embodiment, as shown in FIG. 5, the terminal-side outer locking structure **1155** may have a shape that protrudes outwardly (e.g., in the $-y$ -axis direction) from the support housing body **1151** as the terminal-side outer locking structure **1155** extends in a direction (e.g., the $-z$ -axis direction) where the terminal support housing **115** is separated from the terminal insertion groove **1113**. In this structure, when the terminal support housing **115** is inserted into the terminal insertion groove **1113** by a predetermined length or more, the terminal-side outer locking structure **1155** is caught on the inner wall of the outer case **111**, thereby inhibiting the terminal support housing **115** from being separated from the terminal insertion groove **1113**.

[0055] Referring to FIGS. 7 and 8, the intermediate connector **12** according to an embodiment may include the bidirectional terminal **121**, the device-side intermediate housing **122**, and the connector-side intermediate housing **123**.

[0056] According to an embodiment, the bidirectional terminal **121** may include, for example, a terminal body **1211**, a device-side connection portion **1212** extending in one direction (e.g., the $-z$ -axis direction) from the terminal body **1211** and elastically contactable on both sides of the device bus bar **23**, and a connector-side connection portion **1213** extending in the other direction (e.g., the $+z$ -axis direction) from the terminal body **1211** and elastically contactable on both sides of the terminal bus bar **116**.

[0057] According to an embodiment, the device-side connection portion **1212** may include a pair of legs, and each end of the pair of legs may be formed to protrude inwardly, thereby being in contact with the device bus bar **23**. For example, a portion of the pair of legs excluding the end may have a smaller cross-sectional area than the end. In this shape, when the device bus bar **23** is inserted between the pair of legs, the pair of legs may be easily and elastically deformed to insert the device bus bar **23** smoothly, and in a state where the device bus bar **23** is inserted, each end of the pair of legs may be in close contact with the device bus bar **23** by an elastic restoring force of the pair of legs.

[0058] According to an embodiment, the connector-side connection portion **1213** may include a pair of legs, and each end of the pair of legs may be formed to protrude inwardly, thereby being in contact with the terminal bus bar **116**. For example, unless otherwise stated, the description of the device-side connection portion **1212** may be equally applied to the connector-side connection portion **1213**, and a detailed description thereof is omitted.

[0059] According to an embodiment, the device-side intermediate housing **122** may include a first intermediate housing body **1221**, an accommodation space **1222** in which the bidirectional terminal **121** is accommodated, a device-side connection slot **1223** that communicates with the accommodation space **1222** and through which the device bus bar **23** may pass, and a first locking structure **1224** that is caught on the connector-side intermediate housing **123**.

[0060] According to an embodiment, the first intermediate housing body **1221** may be formed of an insulating material (e.g., resin) to reduce power from being directly transmitted from the bidirectional terminal **121** or the device bus bar **23** to an inner wall of the intermediate groove **22** of the electronic device **2**.

[0061] According to an embodiment, the accommodation space **1222** may be formed to be recessed from an outer surface (e.g., an upper surface of FIG. **8**) of the first intermediate housing body **1221** in which an opening into which the bidirectional terminal **121** is inserted is positioned. For example, the accommodation space **1222** may have a width formed in a direction (e.g., the $+/-x$ -axis direction) parallel to the outer surface (e.g., the left surface with respect to FIG. **3**) of the electronic device **2**. In this shape, the number of the bidirectional terminals **121** to be accommodated in the accommodation space **1222** may be adjusted according to a width of the device bus bar **23** or the size of power flowing through the device bus bar **23**. For example, the bidirectional terminal **121** may be prevented from being separated to the outside by a bottom wall of the accommodation space **1222**.

[0062] According to an embodiment, as shown in FIG. **8**, the device-side connection slot **1223** may be formed to penetrate the bottom wall of the accommodation space **1222** and communicate with the accommodation space **1222**. For example, the device-side connection slot **1223** may be formed to have a size and a shape that allow the device bus bar **23** to pass through and prevent the bidirectional terminal **121** from being separated.

[0063] According to an embodiment, as shown in FIG. **8**, the first locking structure **1224** may be formed to protrude outwardly from the first intermediate housing body **1221** as the first locking structure **1224** extends in a direction (e.g., the $-z$ -axis direction) where the first intermediate housing body **1221** moves away from the second intermediate housing body **1231**. The first locking structure **1224** may reduce the connector-side intermediate housing **123** from being separated from the device-side intermediate housing **122**.

[0064] According to an embodiment, the connector-side intermediate housing **123** may include the

second intermediate housing body **1231**, an insertion space **1232** in which the device-side intermediate housing **122** is inserted, a connector-side connection slot **1233** that communicates with the insertion space **1232** and through which the terminal bus bar **116** may pass, a second locking structure **1234** that is caught on the device-side intermediate housing **122**, a device fastening structure **1235**, and an insulating guide **1236**.

[0065] According to an embodiment, the second intermediate housing body **1231** may be formed of an insulating material (e.g., resin) to reduce power from being directly transmitted from the bidirectional terminal **121** or the terminal bus bar **116** to the inner wall of the intermediate groove **22** of the electronic device **2**.

[0066] According to an embodiment, the insertion space **1232** may be formed to be recessed from an outer surface (e.g., a lower surface of FIG. **8**) of the second intermediate housing body **1231** in which an opening into which the bidirectional terminal **121** is inserted is positioned. For example, the insertion space **1232** may have a width formed in a direction (e.g., the $\pm x$ -axis direction) parallel to the outer surface (e.g., the left surface with respect to FIG. **3**) of the electronic device **2**. For example, the insertion space **1232** may have a cross-sectional shape corresponding to a circumference of the first intermediate housing body **1221**. In this structure, the intermediate connector **12** may be assembled by inserting the device-side intermediate housing **122** into the insertion space **1232** in a state where the bidirectional terminal **121** is inserted into the device-side intermediate housing **122**.

[0067] According to an embodiment, the connector-side connection slot **1233** may be formed to penetrate a bottom wall of the insertion space **1232** and communicate with the insertion space **1232**. For example, the connector-side connection slot **1233** may be formed to have a size and a shape that allow the terminal bus bar **116** to pass through and prevent the bidirectional terminal **121** from being separated.

[0068] According to an embodiment, the second locking structure **1234** may be coupled to the first locking structure **1224** when the first intermediate housing body **1221** is inserted into the second intermediate housing body **1231** by a predetermined distance or more. For example, the second locking structure **1234** may have a groove or hole shape into which the first locking structure **1224** may be inserted. For example, the second locking structure **1234** may have a shape that extends in a direction (e.g., the $-z$ -axis direction) away from the connector-side connection slot **1233** in a cantilever shape from the second intermediate housing body **1231**. For example, the second locking structure **1234** may be elastically deformed outwardly from the second intermediate housing body **1231**. In this structure, in a process of inserting the device-side intermediate housing **122** into the connector-side intermediate housing **123**, the second locking structure **1234** may be opened outward and may pass through the first locking structure **1224**, and in a state where the device-side intermediate housing **122** is inserted into the connector-side intermediate housing **123** by a predetermined length or more, the second locking structure **1234** may be deformed to an original shape by an elastic restoring force and may be caught by the first locking structure **1224**.

[0069] According to an embodiment, the device fastening structure **1235** may be formed to protrude outwardly from the second intermediate housing body **1231** in a direction (e.g., the $\pm z$ -axis direction) perpendicular to a direction (e.g., the $+z$ -axis direction) in which the first intermediate housing body **1221** is inserted into the second intermediate housing body **1231**, as shown in FIG. **7**. For example, in the device fastening structure **1235**, a fastening hole (**1235a**) may be formed through which a fastening structure may pass, wherein the fastening structure is fixable to the device case **21** by being inserted in a direction (e.g., the $-z$ -axis direction) opposite to a direction in which the first intermediate housing body **1221** is inserted into the second intermediate housing body **1231**.

[0070] According to an embodiment, the insulating guide **1236** may have a shape that surrounds the connector-side connection slot **1233** and may be formed to extend from the second intermediate housing body **1231** toward the bent connector **11**. For example, one side of the terminal support

housing **115** may be inserted into an inner side of the insulating guide **1236**. The terminal bus bar **116** may be connected to the connector-side connection portion **1213** by passing through the insulating guide **1236** and the connector-side connection slot **1233** sequentially.

[0071] However, when the distance between adjacent conductive structures is reduced below a predetermined distance, there may be a risk of spark discharge. The distance between two conductive structures causing spark discharge represented as the shortest distance along a surface of a solid dielectric is referred to as a creeping distance. According to the insulating guide **1236** described above, by increasing the creeping distance, electrical stability of the entire connector assembly **1** and the electronic device **2** may be improved.

[0072] For example, in a state where the bidirectional terminal **121**, the device-side intermediate housing **122**, and the connector-side intermediate housing **123** are connected to each other, the intermediate connector **12** may have a shape that surrounds the remaining portion except for the device-side connection slot **1223** and the connector-side connection slot **1233** formed in a longitudinal direction of the bidirectional terminal **121**. For example, the remaining portion of the bidirectional terminal **121** except for the device-side connection slot **1223** and the connector-side connection slot **1233** may not be exposed to the outside. For example, a circumferential direction of the bidirectional terminal **121** may not be exposed to the outside of the intermediate connector **12** by the device-side intermediate housing **122** and/or the connector-side intermediate housing **123**. For example, the device-side connection slot **1223** may be formed only on one surface of the first intermediate housing body **1221** and may not be formed on the remaining surface of the first intermediate housing body **1221**. For example, the connector-side connection slot **1233** may be formed only on one surface of the second intermediate housing body **1231** and may not be formed on the remaining surface of the second intermediate housing body **1231**. In such a shape, the intermediate connector **12** may allow insertion of the device bus bar **23** and the terminal bus bar **116** and may reduce a portion of an internal structure of the intermediate connector **12** exposed to the outside, thereby reducing an issue of foreign substances entering the inside of the intermediate connector **12** from the outside.

[0073] Referring to FIG. **9**, a bent connector **31** (e.g., the bent connector **11** of FIG. **3**) according to an embodiment may include an outer case (e.g., the outer case **111** of FIG. **3**), an inner housing (e.g., the inner housing **112** of FIG. **3**), a fastening nut **313** (e.g., the fastening nut **113** of FIG. **3**), a wire assembly **314** (e.g., the wire assembly **114** of FIG. **3**), a terminal support housing (e.g., the terminal support housing **115** of FIG. **3**), a terminal bus bar **316** (e.g., the terminal bus bar **116** of FIG. **3**), and a fastening bolt **317** (e.g., the fastening bolt **117** of FIG. **3**).

[0074] FIG. **9** illustrates a coupling relationship between the terminal bus bar **316** and the wire assembly **314**. For example, according to the structure illustrated in FIG. **9**, the terminal support housing **115** as shown in FIG. **3** may be omitted, but embodiments are not limited thereto.

[0075] According to an embodiment, the wire assembly **314** may include a conductive wire (e.g., the conductive wire **1141** of FIG. **3**), an inner insulation layer (**3142**) (e.g., the inner insulation layer **1142** of FIG. **3**), an outer insulation layer (e.g., the outer insulation layer **1144** of FIG. **3**), a shield layer (e.g., the shield layer **1143** of FIG. **3**), a grounding ferrule (e.g., the grounding ferrule **1145** of FIG. **3**), a conductive plate **3146** (e.g., the conductive plate **1146** of FIG. **3**), a wire seal (e.g., the wire seal **1147** of FIG. **3**), and a wire-side cover (e.g., the wire-side cover **1148** of FIG. **3**). According to an embodiment, the conductive plate **3146** may include a guide groove **3146a**. According to an embodiment, the guide groove **3146a** may have a shape recessed from at least one side surface of both side surfaces of the conductive plate **3146**, but embodiments are not limited thereto.

[0076] According to an embodiment, the fastening nut **313** may be fixed to the conductive plate **3146**. For example, the fastening nut **313** may be fixed to a hole formed in the conductive plate **3146** in a press fit and/or welding method, but embodiments are not limited thereto.

[0077] According to an embodiment, the terminal bus bar **316** may include a vertical portion **3161**

(e.g., the vertical portion **1161** of FIG. 5), a horizontal portion **3162** (e.g., the horizontal portion **1162** of FIG. 5), and a guide protrusion **3163**. According to an embodiment, the guide protrusion **3163** may have a shape that is cut and bent from a portion of the horizontal portion **3162**. For example, the guide protrusion **3163** may be inserted into the guide groove **3146a** so that the terminal bus bar **316** may be properly arranged with the conductive plate **3146**. In this arranged state, by fastening the fastening bolt **317** to the fastening nut **313**, the conductive plate **3146** and the horizontal portion **3162** may be mechanically and electrically coupled to each other.

[0078] As described above, although the embodiments have been described with reference to the limited drawings, one of ordinary skill in the art may apply various technical modifications and variations based thereon. For example, suitable results may be achieved if the described techniques are performed in a different order and/or if components in a described system, architecture, device, or circuit are combined in a different manner and/or replaced or supplemented by other components or their equivalents.

[0079] Therefore, other implementations, other embodiments, and/or equivalents of the claims are within the scope of the following claims.

Claims

1. A bent connector, comprising: an outer case including a case body, an inner insertion groove recessed in a first direction from a first side of the case body toward an inside of the case body, and a terminal insertion groove recessed in a second direction from a second side of the case body toward the inside of the case body, the second direction intersecting with the first direction; an inner housing formed of an insulating material and inserted through the inner insertion groove from the first side of the case body; a wire assembly having at least a portion inserted into an inside of the inner housing in a direction in which the inner housing is inserted into the inner insertion groove; a terminal bus bar inserted through the terminal insertion groove from the second side of the case body and electrically connected to the wire assembly; and a fastening bolt inserted from the second side of the case body and mechanically and electrically fastening the terminal bus bar and the wire assembly to each other.

2. The bent connector of claim 1, wherein the inner insertion groove and the terminal insertion groove communicate with each other; the inner housing includes: an inner body having at least a portion located where the inner insertion groove and the terminal insertion groove meet each other; a wire insertion groove recessed in the first direction; and a plate insertion groove formed to be further recessed in the first direction from the wire insertion groove, and the wire assembly includes: a conductive wire formed of a conductive material and inserted along the wire insertion groove; and a conductive plate connected to an end of the conductive wire and inserted and supported along the plate insertion groove.

3. The bent connector of claim 2, wherein the wire assembly includes: an inner insulation layer wrapping the conductive wire; an outer insulation layer wrapping the inner insulation layer; a shield layer having at least a portion of the shield layer located between the inner insulation layer and the outer insulation layer and configured to transmit current leaking from the conductive wire to an outside; and a grounding ferrule having a first side in contact with the shield layer and a second side in contact with the case body.

4. The bent connector of claim 3, wherein the grounding ferrule has a contact portion that is formed to protrude toward an inner wall of the case body and is elastically deformable.

5. The bent connector of claim 3, wherein the inner housing has a ferrule insertion groove recessed stepwise from an outer surface of the inner body, and at least a portion of the grounding ferrule is inserted into a space between an inner wall of the case body and the ferrule insertion groove.

6. The bent connector of claim 2, further comprising a fastening nut arranged on an opposite side of the terminal bus bar with respect to the conductive plate, the fastening nut is fastened to the

fastening bolt.

7. The bent connector of claim 6, wherein the inner housing has a nut groove recessed in a direction in which the fastening bolt is inserted into the case body from the plate insertion groove, the fastening nut and the nut groove have polygonal shapes that are complementary to each other.

8. The bent connector of claim 2, wherein the terminal bus bar includes: a vertical portion extending parallel to a longitudinal direction of the terminal insertion groove; and a horizontal portion having a bent shape from the vertical portion, extending parallel to a longitudinal direction of the inner insertion groove, and being in surface contact with the conductive plate.

9. The bent connector of claim 8, wherein a normal direction to a surface of the conductive plate is parallel to the second direction, and a normal direction to a surface of the vertical portion is parallel to the first direction.

10. The bent connector of claim 8, further comprising a terminal support housing inserted through the terminal insertion groove from the second side of the case body and formed of an insulating material, the terminal support housing has a bus bar insertion hole into which the vertical portion is inserted and a bus bar mounting groove in which the horizontal portion is mounted.

11. The bent connector of claim 10, wherein the terminal support housing has a bolting hole that is connected to the bus bar mounting groove and through which the fastening bolt passes.

12. The bent connector of claim 2, wherein a width direction of the conductive plate and a width direction of the terminal bus bar are each parallel to a width direction of the outer case.

13. The bent connector of claim 1, wherein the inner housing has an inner-side outer locking structure having a shape that protrudes outwardly from the inner housing, and when the inner housing is inserted into the inner insertion groove by at least a predetermined length, the inner-side outer locking structure is caught on an inner wall of the outer case, inhibiting separation of the inner housing from the inner insertion groove.

14. An intermediate connector configured to connect an electronic device having a device case and a device bus bar electrically connected to an internal component of the device case to a bent connector including a wire assembly and a terminal bus bar that is electrically connected to the wire assembly and has an end portion in a bent shape with respect to a longitudinal direction of the wire assembly, the intermediate connector comprising: a bidirectional terminal including a terminal body, a device-side connection portion extending in a first direction from the terminal body and elastically contacting a pair of sides of the device bus bar, and a connector-side connection portion extending in a second direction from the terminal body and elastically contacting a pair of sides of the terminal bus bar; a device-side intermediate housing including a first intermediate housing body, an accommodation space in which the bidirectional terminal is accommodated, and a device-side connection slot that communicates with the accommodation space and through which the device bus bar passes; and a connector-side intermediate housing including a second intermediate housing body, an insertion space in which the device-side intermediate housing is inserted, and a connector-side connection slot that communicates with the insertion space and through which the terminal bus bar passes.

15. The intermediate connector of claim 14, wherein the connector-side intermediate housing includes a device fastening structure that protrudes outwardly from the second intermediate housing body in a direction perpendicular to a direction in which the first intermediate housing body is inserted into the second intermediate housing body, and in the device fastening structure, a fastening hole is formed through which a fastening structure passes, the fastening structure is fixable to the device case by being inserted in a direction opposite to the direction in which the first intermediate housing body is inserted into the second intermediate housing body.

16. The intermediate connector of claim 14, wherein the device-side intermediate housing has a first locking structure that protrudes outwardly, and the connector-side intermediate housing has a second locking structure coupled to the first locking structure when the first intermediate housing body is inserted into the second intermediate housing body by at least a predetermined distance, the

second locking structure is elastically deformable in an outward direction of the second intermediate housing body.

17. A connector assembly, comprising: a bent connector including an outer case, a wire assembly inserted from a first side of the outer case, a terminal bus bar inserted from a second side of the outer case, and a fastening bolt inserted from the second side of the outer case and mechanically and electrically fastening the terminal bus bar and the wire assembly to each other; and an intermediate connector installable in an electronic device having a device case and a device bus bar electrically connected to an internal component of the device case, the intermediate connector mechanically and electrically connects the device bus bar to the terminal bus bar.

18. The connector assembly of claim 17, wherein the intermediate connector includes: a bidirectional terminal including a terminal body, a device-side connection portion extending in a first direction from the terminal body and elastically contacting a pair of sides of the device bus bar, and a connector-side connection portion extending in a second direction from the terminal body and elastically contacting a pair of sides of the terminal bus bar; a device-side intermediate housing including a first intermediate housing body, an accommodation space in which the bidirectional terminal is accommodated, and a device-side connection slot that communicates with the accommodation space and through which the device bus bar passes; and a connector-side intermediate housing including a second intermediate housing body, an insertion space in which the device-side intermediate housing is inserted, and a connector-side connection slot that communicates with the insertion space and through which the terminal bus bar passes.

19. The connector assembly of claim 17, wherein a width direction of the terminal bus bar is parallel to a width direction of the outer case.

20. The connector assembly of claim 18, wherein the connector-side intermediate housing has an insulating guide having a shape that surrounds the connector-side connection slot and extending from the second intermediate housing body toward the bent connector.
